

MENTAL HEALTH MONITORING SYSTEM

Phase 3 v2 — Two-Stage End-to-End Classifier Training

Progress Report

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Platform: Kaggle Notebooks | Date: July 25, 2026

1. Project Overview

The Mental Health Monitoring System is a multi-phase NLP project designed to automatically detect depression, classify mental health severity, and generate clinically coherent reports from raw text input. The system is structured as a four-layer architecture:

- Layer 1 — Transformer backbone (MentalRoBERTa) for contextual text embeddings
- Layer 2 — Graph Attention Network (GAT) for structural reasoning over token dependency graphs
- Layer 3 — Three-head classifier: emotion recognition, mental health category, and severity regression
- Layer 4 — Large Language Model (Groq) for report generation and crisis alerting

This report covers a revised version of Phase 3 ("v2"). The architecture and dataset are unchanged from the first Phase 3 notebook, but the training procedure has been redesigned around a two-stage fine-tuning schedule instead of single-stage joint training from epoch one.

2. Phase 3 v2 Objective

Retrain the same end-to-end model — MentalRoBERTa backbone, GAT, and emotion/MH-category heads — using a two-stage warm-up strategy: first stabilize the GAT and classification heads with the backbone frozen, then unfreeze the backbone for full joint fine-tuning with early stopping. The goal is to reduce the risk of the pretrained backbone being disrupted by large, noisy gradients from randomly-initialized heads early in training, and to see whether this improves on the first version's results — particularly on the rare emotion classes it struggled with.

2.1 What Changed From the First Phase 3 Version

The dataset, split sizes, tokenizer, batch sizes, and model architecture are all identical to the first version. Three things changed:

- **Two-stage training schedule:** Stage 1 trains only the GAT and heads for 5 epochs with the backbone frozen ($\text{lr} = 5\text{e-}4$). Stage 2 reloads the best Stage 1 checkpoint, unfreezes the backbone, and continues training for up to 15 epochs with differential learning rates (backbone $2\text{e-}5$, GAT/heads $5\text{e-}5$ — the same rates the first version used from epoch one) and early stopping (patience = 5 on validation macro-F1).
- **Explicit GAT weight transfer:** instead of automatically matching all keys in the Phase 2 checkpoint by shape, this version defines an explicit `key_map` naming exactly the 8 GAT parameter tensors (`att_src`, `att_dst`, `bias`, `lin.weight` for `gat1` and `gat2`) that should transfer. All 8 matched. This is functionally equivalent to the first version's result (which also matched 8, alongside 2 keys that were never meant to transfer) but states the intent directly in code rather than relying on automatic shape-matching to happen to exclude the old Phase 2 classifier head.
- **Inference-time "top emotion" view:** after evaluation, the notebook additionally prints just the single highest-confidence emotion (and its score) for 5 sample comments, instead of showing probabilities across all 28 classes. This is a presentation aid for demoing the model one comment at a time — it does not affect training, metrics, or the saved checkpoint.

3. Environment and Configuration

Platform	Kaggle Notebooks
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Hardware	GPU (CUDA)
Frameworks	PyTorch, PyTorch Geometric (GATConv), HuggingFace Transformers, datasets, accelerate
Phase 1 checkpoint	mentalroberta_phase1_final — located dynamically via glob, found successfully
Phase 2 checkpoint	gat_phase2_best.pt — located dynamically via glob, found successfully
Max Sequence Length	128 tokens

4. Dataset — GoEmotions (Unchanged)

Source	3 raw CSV files (goemotions_1/2/3.csv), combined: 211,225 rows × 37 columns
Cleaning	Unclear examples removed; deduplicated via groupby("id") + max() across raters per emotion
After deduplication	58,009 unique comments × 28 emotion labels
Train / Val / Test	49,307 / 4,351 / 4,351 (85% / 7.5% / 7.5%), random_state = 42

Identical split sizes to the first Phase 3 version confirm the data pipeline itself is unchanged — only the training procedure differs from here on.

5. Model Architecture (Unchanged)

Component	Description
Layer 1 — Backbone	MentalRoBERTa loaded from the Phase 1 checkpoint; produces 768-dim token embeddings
Layer 2 — GAT	gat1: 768 → 4 heads × 256 = 1024 (concatenated); gat2: 1024 → 2 heads → 256 (averaged). Sequential token-to-token edges, cached per (batch, seq_len)
Layer 3 — Emotion head	256 → 128 → 28, multi-label (sigmoid applied in loss/inference)
Layer 3 — MH category head	256 → 64 → 2, binary softmax classification
Layer 3 — Severity head	256 → 64 → 1, regression with sigmoid output; frozen, pending DAIC-WOZ access
Pooling	CLS-token embedding taken as the graph-level vector after both GAT layers

6. Transfer of Phase 2 GAT Weights

Layers matched	8 / 8 — explicit key map (att_src, att_dst, bias, lin.weight × gat1/gat2)
Old Phase 2 classifier head	Not referenced at all — cleanly excluded by the key map rather than filtered out after an automatic shape-match

7. Training Strategy — Two Stages

7.1 Stage 1 — Backbone Frozen (5 Epochs)

Only the GAT and the emotion/MH heads are trained ($lr = 5e-4$); the backbone is fully frozen so it cannot be disturbed by large early gradients from the randomly-initialized heads.

Epoch	Train Loss	Val F1	Best?
1	0.1192	0.4902	✓
2	0.1138	0.4882	
3	0.1126	0.4897	
4	0.1118	0.5044	✓
5	0.1112	0.5071	✓ (best)

Stage 1 alone reaches 0.5071 validation macro-F1 — already higher than the entire 3-epoch single-stage run from the first Phase 3 version (0.5008) — with the backbone still completely frozen.

7.2 Stage 2 — Backbone Unfrozen (Up to 15 Epochs, Early Stopping)

The Stage 1 best checkpoint is reloaded, the backbone is unfrozen, and training continues at differential learning rates (backbone $2e-5$, GAT/heads $5e-5$) with early stopping (patience = 5).

Epoch	Train Loss	Val F1	Note
1	0.1132	0.5033	best
2	0.1071	0.5118	best
3	0.1008	0.5055	no improve 1/5
4	0.0946	0.5121	best (final checkpoint)
5	0.0883	0.5002	no improve 1/5
6	0.0819	0.5007	no improve 2/5
7	0.0757	0.4999	no improve 3/5
8	0.0698	0.4967	no improve 4/5
9	0.0641	0.4920	no improve 5/5 → stopped

Training loss keeps falling every epoch, but validation F1 peaks at epoch 4 (0.5121) and steadily declines afterward — a clear overfitting pattern once the backbone starts moving. Early stopping correctly halts training at epoch 9 and the epoch-4 checkpoint is kept. Overall improvement across both stages: 0.5071 → 0.5121.

8. Final Test Evaluation

8.1 Overall Metrics

Macro F1 (all 28 classes equally weighted)	0.51
Micro F1 (weighted by class frequency)	0.59
Samples F1 (per-sample average)	0.61

8.2 Per-Emotion Breakdown

Emotion	Precision	Recall	F1	Support
admiration	0.67	0.71	0.69	764
amusement	0.82	0.71	0.76	408
anger	0.49	0.60	0.54	417
annoyance	0.44	0.59	0.50	712
approval	0.50	0.52	0.51	990
caring	0.49	0.47	0.48	343
confusion	0.54	0.59	0.57	396
curiosity	0.63	0.77	0.69	432
desire	0.44	0.46	0.45	181
disappointment	0.40	0.46	0.43	490
disapproval	0.43	0.68	0.53	615
disgust	0.42	0.53	0.47	303
embarrassment	0.44	0.27	0.34	132
excitement	0.36	0.46	0.40	318
fear	0.54	0.58	0.56	167
gratitude	0.74	0.74	0.74	385
grief	0.33	0.08	0.13	49
joy	0.50	0.51	0.51	433
love	0.82	0.76	0.79	335
nervousness	0.41	0.27	0.32	119
optimism	0.55	0.43	0.49	482
pride	0.50	0.13	0.21	99
realization	0.30	0.28	0.29	529
relief	0.37	0.23	0.28	92
remorse	0.72	0.58	0.64	127
sadness	0.54	0.58	0.55	365
surprise	0.50	0.47	0.49	267
neutral	0.71	0.87	0.78	2,352
Macro avg	0.52	0.51	0.51	12,302
Micro avg	0.56	0.61	0.59	12,302
Weighted avg	0.56	0.61	0.58	12,302

Emotion	Precision	Recall	F1	Support
Samples avg	0.62	0.68	0.61	12,302

8.3 Interpretation

The overall pattern is similar to the first version — common emotions perform best, rare ones worst — but rare-class recall is meaningfully better this time: pride's F1 rose from 0.02 to 0.21, grief from 0.08 to 0.13, and relief from 0.16 to 0.28. This comes at a small precision cost on several mid-frequency classes (e.g. disapproval precision fell from 0.47 to 0.43 while its recall rose from 0.58 to 0.68; realization dropped slightly from 0.31 to 0.29 F1). Overall, the two-stage schedule appears to nudge the model away from the extremely conservative, high-precision/near-zero-recall behaviour the first version showed on rare classes, toward a more balanced — if still imperfect — trade-off.

9. Key Outputs and Deliverables

Deliverable	Location	Purpose
phase3v2_s1_best.pt	/kaggle/working/	Best Stage 1 checkpoint (backbone frozen), used as the Stage 2 starting point
phase3v2_best.pt	/kaggle/working/	Best Stage 2 checkpoint (epoch 4, val F1 0.5121), used for final test evaluation
phase3v2_final.pt	/kaggle/working/	Final saved model state after test evaluation
phase3v2_tokenizer/	/kaggle/working/	Tokenizer saved alongside the model for downstream reuse

10. Comparison — v1 vs. v2

	v1 (Single-Stage)	v2 (Two-Stage)
Training schedule	3 epochs, everything trained jointly from epoch 1	Stage 1: 5 epochs, backbone frozen. Stage 2: up to 15 epochs, unfrozen, early stopping
Total epochs run	3	14 (5 + 9, stopped early)
Best validation macro-F1	0.5008	0.5121
Test macro-F1	0.49	0.51
Test micro-F1 / samples-F1	0.59 / 0.61	0.59 / 0.61 (unchanged)
Rare-class recall (grief, pride, relief)	Near zero	Noticeably better, still weak
GAT weight transfer	8/10 matched (auto shape-match)	8/8 matched (explicit key map)
Overfitting visible?	Not visible — run too short to show it	Clearly visible — Stage 2 val F1 peaks at epoch 4, degrades for 5 straight epochs
Approx. training time	~1 hour (3 × ~20 min/epoch)	~4–5 hours (14 × ~20 min/epoch)

11. Phase 3 v2 Summary

Metric	Value	Notes
Training samples (GoEmotions)	49,307	85% of 58,009 deduplicated comments
Train / Val / Test split	49,307 / 4,351 / 4,351	85% / 7.5% / 7.5% — unchanged from v1
Phase 2 GAT weights transferred	8 / 8 layers	Explicit key map; old PRIMATE head cleanly excluded
Best validation macro-F1	0.5121	Stage 2, epoch 4
Test macro-F1	0.51	Up from 0.49 in v1; still weak on rare classes
Test micro-F1 / samples-F1	0.59 / 0.61	Unchanged from v1
Severity head	Frozen	Included in architecture; untrained pending DAIC-WOZ access

12. Next Steps — Phase 4 and Remaining Phase 3 Work

Phase 3 v2 is functionally complete for the emotion and MH-category heads, with a modest improvement over the first version. The severity head — and full validation of the joint model — still depends on data pending approval.

- Submit / follow up on the DAIC-WOZ data use agreement if not already approved — this blocks unfreezing and training the severity head
- Once DAIC-WOZ access is granted, unfreeze the severity head and fine-tune it jointly with the rest of the pipeline on PHQ-8 severity scores
- Consider tightening Stage 2's early-stopping patience (e.g. 2–3 instead of 5) to reduce wasted training time without sacrificing the accuracy gain
- Revisit the sequential-edge vs. dependency-edge discrepancy (both versions use simple sequential edges rather than Phase 2's spaCy dependency edges) — consider whether a differentiable or precomputed dependency structure could be reintroduced
- Continue investigating recall on rare emotions (grief, pride, relief, nervousness) — e.g. targeted oversampling or per-class decision thresholds
- Replace the synthetic MH-category proxy label with a more clinically grounded signal once a suitable labelled dataset is available
- Begin scoping Phase 4 — Groq LLM integration for report generation and crisis alerting — once the full three-head model (including severity) is validated

13. Phase Status

Phase	Status	Notes
Phase 1 — Backbone Fine-Tuning	✓ COMPLETE	Checkpoint committed to Kaggle Output
Phase 2 — GAT Training	✓ COMPLETE	gat_phase2_final.pt saved; test macro-F1 0.5234

Phase	Status	Notes
Phase 3 — Three-Head Training	IN PROGRESS	v2 (two-stage) recommended over v1: macro-F1 0.51 vs 0.49; severity head frozen pending DAIC-WOZ
Phase 4 — LLM Integration	PENDING	Groq API integration — awaiting Phase 3 completion